FIKR Model — Lesson Plan

Learning Leading to Production

Title	Identities with Polynomials	**Subject / Unit**	Topic 3 — Polynomial Functions · Lesson 3-3
Grade	Grade 10 — Algebra II	**Date**	Week 10 · Semester 1, 2026–27
Format	[ ] 40-minute [X] 60-minute	**Teacher**	Mr Malek Thiab

Essential Question (curriculum map)

Not shown — the Scope and Sequence document has no Essential Question column, so it is left open rather than invented. Add it when confirmed.

Lesson Objectives (curriculum map — verbatim)

1. Understand and prove polynomial identities.
2. Apply polynomial identities for multiplication, factoring, and simplification.
3. Expand a power of a binomial by applying the binomial theorem.

Standards (curriculum map — verbatim)

HSA.SSE.A.2 Use the structure of an expression to identify ways to rewrite it.

HSA.APR.C.4 Prove polynomial identities and use them to describe numerical relationships.

(+)HSA.APR.C.5 Know and apply the Binomial Theorem for the expansion of (x + y)n in powers of x and y for a positive integer n, where x and y are any numbers, with coefficients determined for example by Pascal's Triangle.

Vocabulary (curriculum map) · Assessments (curriculum map)

Vocabulary: Binomial Coefficient ; binomial theorem ; Identity ; Pascal's triangle.

Assessments: "Give it a go" questions during practice ; "Time to Check" at the Mastery Gate.

0 Preparation — Before Class

Learning Outcome: What will the student be able to do?

Students will understand and prove a polynomial identity by expanding one side until it matches the other (objective 1; HSA.APR.C.4).

Students will apply $(a+b)^2$, $(a-b)^2$ and $(a+b)(a-b)$ to expand, factor and simplify, and use them to calculate numbers quickly (objective 2; HSA.SSE.A.2).

Students will expand a power of a binomial using Pascal's triangle and the binomial theorem (objective 3; (+)HSA.APR.C.5).

Students will use identities to describe numerical relationships, such as $99 \times 101 = 100^2 - 1$ and Pythagorean triples (HSA.APR.C.4).

Three teaching days this week. If time runs short, cut Smart Production — never the Mastery Gate.

Pre-Class Preparation — Topic/Link and Evidence of Readiness: What will the student complete before class?

Pre-session home journey (already issued, with Lesson 3-4): watch "How to Use the Binomial Theorem", study two worked examples, answer three ungraded readiness questions. Submitted on the LMS before class.

Readiness questions: expand $(x+1)^4$ from row 4 of Pascal's triangle; evaluate $101^2 - 99^2$ with an identity; expand $(x-2)^3$.

Evidence: the LMS answers feed the Diagnose.

1 Diagnostic 6 min

Tool Used: What evidence indicates differences in students' current levels?

3-question Quizizz (≤ 4 minutes) — the slide-4 checks: expand $(x+4)(x-4)$; simplify $(3x)^2$; evaluate $(a+b)^2$ and $a^2 + b^2$ for $a = 3$, $b = 2$.

Answers: $x^2 - 16$; $9x^2$; 25 and 13.

Where is the learning gap? What will you adjust in your instruction accordingly?

Expected gaps — Q2: $(3x)^2$ written as $3x^2$. Q3: the two values are treated as equal, which is the whole point of the lesson.

Routing: 0–1 correct → re-teach expanding two brackets and squaring a product. 2 → straight to the three square patterns. 3 → begin Investigate early.

2 Targeted Instruction 18 min

Required Instruction and Modeling: What needs direct explanation? Which example will demonstrate the thinking process and the expected quality standard?

Slide 5 — Proving an identity (objective 1; HSA.APR.C.4). $(x+3)^2 = x^2 + 6x + 9$ is proved by expanding one side; $(x+2)^2 = x^2 + 4$ is disproved by one value, $x = 1$: $9 \neq 5$.

Slide 6 — Identities for multiplication and factoring (objective 2; HSA.SSE.A.2). $(2x-5)^2 = 4x^2 - 20x + 25$; $9x^2 - 49 = (3x-7)(3x+7)$; $x^3 - 27 = (x-3)(x^2+3x+9)$.

Slide 7 — Identities and numbers (HSA.APR.C.4). $48^2 = (50-2)^2 = 2304$; $99 \times 101 = 100^2 - 1 = 9999$; $x = 3$, $y = 2$ gives the triple $5,\ 12,\ 13$.

Slide 8 — Binomial theorem and Pascal's triangle (objective 3; (+)HSA.APR.C.5). $(x+2)^4 = x^4 + 8x^3 + 24x^2 + 32x + 16$; $(2x-1)^3 = 8x^3 - 12x^2 + 6x - 1$; the x^2 coefficient of $(x+3)^5$ is 270.

Narration focus — never lose the middle term; the powers of the first term fall while the powers of the second rise.

Quick Check for Understanding: What question or evidence confirms that students are ready to move on to practice?

On whiteboards, 60 seconds: expand $(2x+3)^2$.

Expected: $4x^2+12x+9$. Watch for $4x^2+9$ (middle term lost) and $2x^2+12x+9$ (only x squared).

80% correct → release guided practice.

3 Practice 12 min

Guided Task: How will the student progress from guided to independent practice? What feedback will the student use?

Two identical "Give it a go" problems, worked with the teacher:

(1) Expand $(x+5)^2$ and factor x^2-64. (2) Use row 3 of Pascal's triangle to expand $(x+1)^3$.

Answers: (1) $x^2+10x+25$; $(x-8)(x+8)$. (2) x^3+3x^2+3x+1.

Worked live; students self-mark and circle their OWN first error.

Independent Task and Feedback Tool: students choose their route and may switch at any time

"Choose Your Route" sheet with live Pear Deck feedback. Routes name the task, not the student.

PRACTICE — Secure the method. Six cards: expand, factor and use a row of Pascal's triangle; then find the error in a squared bracket and prove one identity. Done when: the middle term $2ab$ is written, its sign follows the bracket, and each answer is checked with a number.

APPLY — Use it in context. A solar-panel field, an event hall, a divisibility proof and $(x-2)^5$. Done when: the identity used is named and each answer says what it means.

INVESTIGATE — Find out why. Prove $(a+b)^2-(a-b)^2=4ab$, build a Pythagorean triple, and explain the row sums 2^n. Done when: the argument starts from a known identity or a full expansion.

Page 3: compare three expansion routes, judge three claims, test a What if...? idea. Every student sits the same Mastery Gate.

4 Production — Stage One 12 min

Production Task: What new task, situation, or product will allow the student to demonstrate learning?

Non-routine transfer task — this context has not appeared in practice (slide 11).

A model square plaza in the Wadi Hanifa restoration has side x m. A uniform paved walkway 3 m wide runs all round the outside. The model values are invented for practice.

(a) Write the side and area of the plaza with its walkway and expand. (b) Write the walkway area as a difference of squares and factor it. (c) For $x=24$ find the walkway area from the factored form and check it directly.

Answers: (a) side $x+6$, area $x^2+12x+36$. (b) $(x+6)^2-x^2=12x+36=12(x+3)$. (c) $12\times27=324$ m^2; check $30^2-24^2=900-576=324$.

Students then hand their reasoning to the AI challenger.

Product Quality Criteria and the Role of AI: What is the minimum acceptable standard? What must the student complete independently?

Minimum standard: the identity named, every step shown, the factored form used for the number, units (m, m ^{2}).

Role of AI: challenger only — "Challenge my reasoning and point out any step I have not justified." Never "solve this".

Completed independently: the written justification for (a)–(c).

5 Presentation — Evidence of Learning 6 min

Evidence of Learning and Success Criterion: What will the student present or explain, and how? What must be clearly demonstrated for learning to be considered achieved?

"Time to Check" — one individual task. No notes, no partner, no AI.

1. Expand $(4x-3)^2$. 2. Factor $49x^2-81$. 3. Expand $(x+1)^5$ with Pascal's triangle.

Answers: 1. $16x^2-24x+9$. 2. $(7x-9)(7x+9)$. 3. $x^5+5x^4+10x^3+10x^2+5x+1$.

Done when: question 1 keeps the middle term and question 3 uses row 5.

Scored live in Pear Deck and logged in the Impact Dashboard.

6 Final Product — Refinement 6 min

Refinement and Final Product: What revision will the student make based on feedback? What will the final product look like?

Students revise their plaza reasoning using the AI challenger's feedback and the teacher's micro-conference note.

Final product (Canva one-pager, photographed solution, or 45-second voice note): the identity named, the proof, ONE non-example ($(a+b)^2=a^2+b^2$ disproved by a number), and why the walkway area factors as $12(x+3)$.

Graded on clarity of reasoning.

Presentation/Publication and Reflection: Where will the product be presented or published? What reflection question will close the learning cycle?

Published to the class LMS portfolio; the strongest go on the Mathematics Department wall.

Total minutes across the seven stages: 60 (6 + 18 + 12 + 12 + 6 + 6) — matches the 60-minute format, including transitions.

★ Required Pillars Differentiation · Adaptive Learning · Saudi Connection · Exam Alignment

Differentiation — three routes named for the task; students choose, and may switch at any time

PRACTICE — Secure the method. Six cards, a find-the-error line and a proof box. Done when: the middle term is written and each answer is checked with a number.

APPLY — Use it in context. A solar field, a seating count, a divisibility proof and a fifth power. Done when: the identity is named.

INVESTIGATE — Find out why. A proof, a Pythagorean triple and the row sums. Done when: the argument starts from a known identity.

No student is labelled; every student sits the same Mastery Gate.

Adaptive Learning — how the lesson branches on evidence

Entry branch — the Quizizz result routes each student to the re-teach table, straight to the new content, or the Investigate prompt.

Mid-lesson branch — Pear Deck flags error patterns during independent practice; flagged students get a 2-minute micro-conference while others continue.

Exit branch — the Mastery Gate routes each student: PASS → Enrichment & Challenge; NOT YET → Targeted Learning Clinic at the start of the next lesson.

Routing is communicated privately through the LMS, never read aloud.

Saudi Culture & Real-World Connection

Wadi Hanifa plaza, a solar-panel field and an event hall; invented values.

SAT / SAAT (Tahsili) / GAT (Qudurat) Connection

SAT — Advanced Math: if $a - b = 3$ and $ab = 10$, then $a^2 + b^2 = 29$. Trap: answering $9 + 10 = 19$.

GAT (Qudurat) — 49×51 versus 2500: $49 \times 51 = 2499$, so 2500 is greater. Trap: assuming they are equal.

SAAT (Tahsili) — the coefficient of x in $(x + 3)^4$ is 108. Trap: using $4 \times 3 = 12$.

Suggestions for Improvement: